

Data Manipulation with Pandas

Objectives

- Use Jupyter Notebooks
- Use the Pandas library to load data
- Identify and handle missing data

Why use Pandas for Data Manipulation

This lesson will give you the tools to be able to load, clean, and better understand your data. This is an important topic as you can't apply machine learning algorithms to data that isn't clean and in a format that an algorithm can understand. For instance, most algorithms can't have missing data, need specific column types, and require a minimum number of rows.

Exercises

Jupyter Notebooks and Datasets

Download and unzip this zip file: [Notebooks_Pandas.zip](#)

(<https://vetsintech.instructure.com/courses/241/files/45308?wrap=1>) 

(https://vetsintech.instructure.com/courses/241/files/45308/download?download_frd=1)

I highly recommend that you have your own copy of each notebook running to follow along with the lesson. [There is also a YouTube video on these notebooks which you can reference as well](#) 

(<https://youtu.be/3qdzsvlOIS4>). In order, the notebooks used are:

1. IntroPandas.ipynb
2. CreateData.ipynb
3. LoadData.ipynb
4. BasicOperations.ipynb
5. Slicing.ipynb
6. Filtering.ipynb
7. RenamingDeletingColumns.ipynb
8. AggregateFunctions.ipynb
9. IdentifyingMissingData.ipynb
10. RemoveFillMissingData.ipynb
11. ConvertNumPyArrayDict.ipynb
12. ExportCSVExcel.ipynb

In case you are curious about how the dataset used for this first video was created, here is a link to learn more: [The Cost of Financing a Car \(Car Loans\)](https://medium.com/data-science/the-cost-of-financing-a-new-car-car-loans-c00997f1aee)  [\(https://medium.com/data-science/the-cost-of-financing-a-new-car-car-loans-c00997f1aee\)](https://medium.com/data-science/the-cost-of-financing-a-new-car-car-loans-c00997f1aee). This concept is applicable for mortgages as well.

Recommended Reading

[Comparing Spark, Dask, Ray, and etc](https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework)  [\(https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework\)](https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework)  [\(https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework\)](https://blog.dominodatalab.com/spark-dask-ray-choosing-the-right-framework)